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A

ASD & D

Assignment - 2

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Q1] Develop a pipeline - CI/CD setup using GitHub Actions or Jenkins

CI/CD means continuous Integration and continuous Delivery/Deployment. It automates software building, testing and deployment. GitHub Actions is a GitHub tool used to automate CI/CD workflows whenever code is pushed or updated.

Pipline

Code → GitHub → Build → Test → Deploy

Stages

- Build : Prepare the application
- Test : Check for errors
- Deploy : Release the application.

Benefits

- Saves Time
- Reduces errors
- Faster Releases
- Improves software quality.

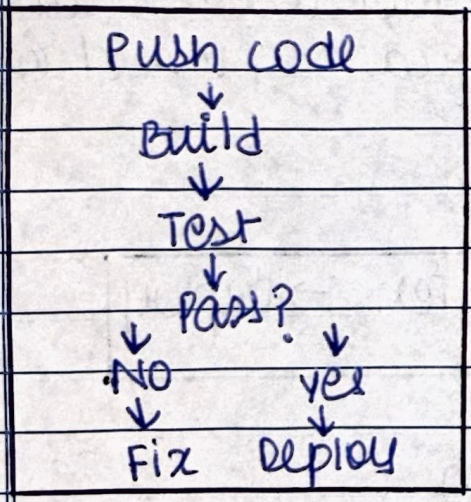
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GitHub Actions Workflow

A workflow is a set of automated instructions that GitHub Actions follows to build, test & deploy an application.

Steps

1) Create a GitHub repository
2) Add the project code
3) Create a workflow file
4) Set a trigger such as push
5) Build & Test the code
6) Deploy if the tests pass



Simple Workflow

Advantages

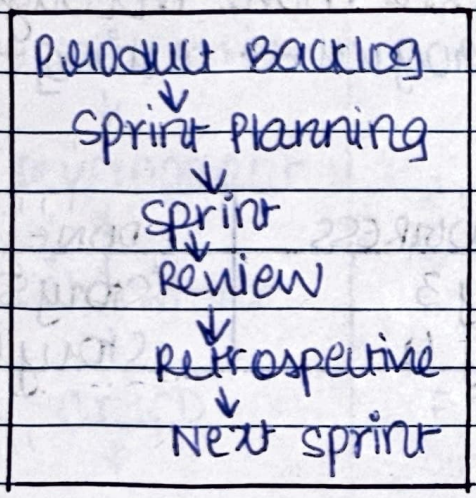
- Automatic Testing
- Faster Deployment
- Less Manual Work
- Early error detection
- Reliable software delivery

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Q2] Agile Process using JIRA

Agile is a software development approach based on short development cycles, teamwork & continuous feedback.

JIRA is a project management tool used to plan & track Agile projects.



Agile cycle.

Main Roles:-

Product Owner - Manages & Prioritises requirements.
Scrum Master - Manages the Scrum Process.
Development Team - Develops & tests the Product.

Product Backlog

It contains all the features and requirements of the project.

These Requirements are written as User Stories

eg] As a customer, I want to log in so that I can access my account.

User stories can be divided into smaller tasks and assigned to team members.

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Agile Implementation in JIRA.

1) Create Backlog - Add all project requirements as User Stories
2) Sprint Planning - Select important stories from the Product Backlog & add to the sprint.
3) Sprint execution - Tasks move through:-
   To do → In progress → Testing → Done.

JIRA BOARD

To DO	IN PROGRESS	DONE
Story 1	Story 3	Story 5
Story 2	Story 4	Story 6.

Daily Scrum
Team Members Discuss:
- What was completed?
- What will be done today?
- Any blockers?

Sprint Review
The completed work is demonstrated to the Product owner and Stakeholders.

Retrospective
The Team discusses what went well, what went wrong, & what can be improved.

JIRA helps plan, track and manage Agile projects efficiently.

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03] Agile Planning with DevOps

Agile focuses on iterative development and customer feedback, while DevOps focuses on automation, testing, & deployment.

Planning
↓
User stories
↓
Development
↓
Git / GitHub
↓
CI / CD
↓
Testing
↓
Deployment
↓
Feedback

Agile + DevOps Flow

Agile decides what to develop, while DevOps focuses to build, test & deliver it efficiently.

This integration results in faster releases, better delivery, & improved software quality.

Integration :-

1) Requirements are created as user stories
2) Stories are selected for a sprint
3) Developers write & commit code using Git.
4) CI / CD automatically build & test the code.
5) Successfully tested code is deployed.

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DevOps Automation

DevOps uses automation to reduce manual work and deliver software quickly.

Main Automation Steps

Version control
[Git / GitHub stores & manages source code]
↓
Continuous Integration
[New code is automatically built & tested]
↓
Automated Testing.
[Test & identify bugs before deployment]
↓
Deployment
[Successfully tested code is automatically deployed]
↓
Monitoring
[The application is monitored after deployment]

Benefits

- Faster software delivery
- Fewer errors
- Better collaboration
- Continuous Testing
- Reliable deployment
- Quick Bug detection.

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example:-
For an online shopping application, a new payment feature is added as JIRA user story. After development, the code is pushed to GitHub, automatically tested and then deployed.

Agile & devops work together to provide faster, automated & reliable software delivery.